Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **Java** and Python backend services, mobile app backends for **Android** and iOS clients, SQL and SQLite data layers, **Docker** deployments, and machine learning systems, from prototype to production.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built backend services and integration pipelines across PostgreSQL, SQL Server, and **distributed cloud storage** with reconciliation checks, sustaining 99%+ reliability over 5K+ monthly jobs through CI/CD releases.
- Restructured fault-tolerant asynchronous LLM job processing with explicit states and failure isolation, tripling throughput across 1K+ jobs, and tuned **relational** schemas to cut dashboard latency from 1.2s to 350ms.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Deployed an AI product end to end, from React UI and REST APIs to MySQL on AWS, integrating **computer vision** OCR and multimodal LLMs for 150+ instructors and 10K+ daily requests.
- Monitored and improved **reliability** by adding vision-model fallbacks, timeout handling, and multithreaded batch processing for 1K+ files per run, cutting feedback turnaround 60% at 90%+ grading consistency.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a GPT-based Decision Transformer for **offline reinforcement learning** in autonomous driving, with pipelines over 100K+ trajectory logs, improving long-horizon trajectory planning 18% in closed-loop **simulation** experiments.
- Designed a multi-head attention encoder fusing HD-map geometry with ego-agent trajectories for **autonomous** planning, benchmarking against three baselines and improving closed-loop planning stability 15% across simulated driving scenarios.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Built object-oriented **Java** and Python modules with Factory and Singleton patterns for cross-platform data-sharing microservices, tested with 20+ Mockito and PyTest suites in nightly Jenkins pipelines across operating systems.
- Troubleshot large-scale API integrations and optimized payloads for **low-bandwidth** clients with pagination, field filtering, and caching, improving response times 30% while working with Microsoft and GitHub stakeholders.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built Spring Boot and Flask backends for a **React Native Android** and iOS booking app, with Firebase real-time sync and push notifications, keeping responses under 200ms across 50K+ records.
- Resolved **mobile** wallet connectivity issues and added wallet-signature login on Android, improving cross-platform compatibility 15% across Android, iOS, and web, and presented the end-to-end architecture to the client before implementation.

Skills

Languages: Java, Python, JavaScript, TypeScript, SQL, C++
Mobile & Backend: React Native, Android, Firebase, Spring Boot, FastAPI, Flask, REST APIs, Microservices
Data: SQLite, PostgreSQL, MySQL, SQL Server, MongoDB, AWS S3, Azure Blob Storage, Hyperparameter Tuning
ML: PyTorch, Reinforcement Learning, Transformers, Computer Vision, LLM Systems, Simulation, Feature Engineering
Infrastructure & Tools: Docker, Kubernetes, AWS, Azure, Git, GitHub Actions, Jenkins, CI/CD, Mockito, PyTest, Linux

Projects

Offline Field Data Capture App | *Kotlin, Jetpack Compose, ObjectBox, WorkManager, Retrofit*

- Designed a native **Kotlin** Android app that records field observations and photos offline in an ObjectBox store, then syncs them to a REST backend with conflict handling once **connectivity** returns.

On-Device Vision Inspection Demo | *Kotlin, TensorFlow Lite, CameraX, Docker, FastAPI*

- Designed an Android **prototype** that runs a TensorFlow Lite detector on the camera feed, flags objects of interest with confidence scores, and uploads results to a containerized FastAPI service for review.

CaptionCraft Live | *Transformers, PyTorch, Flask, Streamlit, Gradio*

- Benchmarked BLIP, GIT, and ViT-GPT-2 **computer vision** models on a 10K-image, 100K-caption dataset, raising caption accuracy from 81% to 96% and serving results through real-time inference APIs for users.